

ELE3305
Design Assignment
Report

Name : _P.Pooja

Date : 03/05/2025

Course : Electronics and Communication Engineering

Introduction :

Designing a distributed system that is a component of an automobile dashboard . My work focuses on requirements, conceptual design, and coding in relation to predetermined hardware.

Ultrasonic parking sensors and fuel usage monitoring sensors are found in the majority of contemporary automobiles. Mainly three units :

- a distance sensor unit,
- a fuel and speed sensor unit, and
- a display unit

each with a single microcontroller make up the system used in this exercise. Automotive environments typically employ the CAN bus. RS485 and MODBUS are the most common in automation. The controllers in this example are linked together using a straight forward RS232 serial interface.

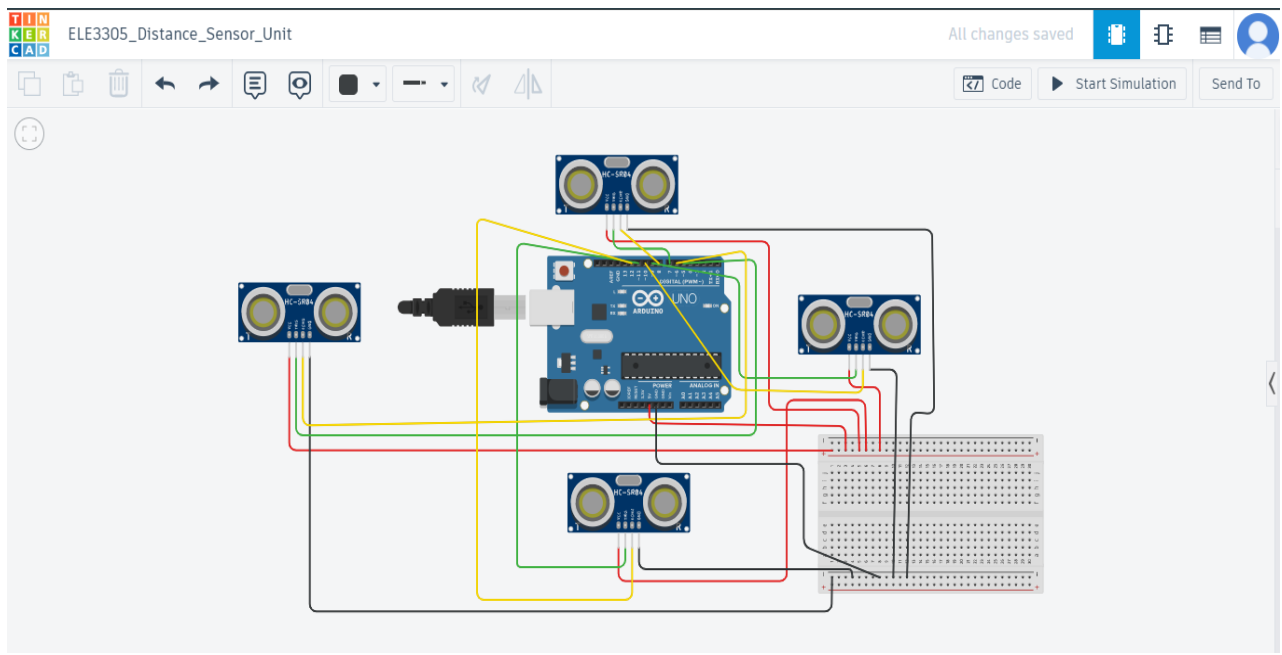
UNIT 1 – Distance Sensor Unit :

Receiving the measured distance values from the sensor—which may come from the Distance Sensor Unit—and giving the user the necessary feedback are the key objectives of the Display Unit. After processing the incoming data, this unit shows the distance value on a LCD screen. The system instantly alerts the user with a warning message on the display if the distance value shows that an obstacle is too close to it, say less than 20 cm.

This device not only sounds an audible warning but also sounds a buzzer in addition to the visual notice. It's crucial to note that the buzzer beeps at intervals that correspond to the obstacle's distance rather than at random. There is a sense of urgency since the beeps are spaced out more quickly the closer the thing is. A mapping function is used to implement this proportional beeping, altering the delay according to the distance measured. With its combination of

visual and aural outputs, the Display Unit serves as the project's main alarm system, making it safer and easier to use. It makes sure the user can swiftly react to prevent danger and is always aware of adjacent obstacles.

Circuit Diagram:



Output:

The screenshot shows the Tinkercad IDE with the code for the Distance Sensor Unit. The code defines four sensors (TRIG0, ECHO0, TRIG1, ECHO1, TRIG2, ECHO2, TRIG3, ECHO3) and a function to read the distance from a sensor. The Serial Monitor shows the output of the function, which is 'D:2' for each sensor. The title bar shows 'ELE3305_Distance_Sensor_Unit' and the status 'All changes saved'. The simulator time is 00:00:15. The code editor shows the following code:

```
// Distance Sensor Unit
1
2
3 #define TRIG0 5
4 #define ECHO0 6
5 #define TRIG1 7
6 #define ECHO1 8
7 #define TRIG2 9
8 #define ECHO2 10
9 #define TRIG3 11
10 #define ECHO3 12
11
12 long readDistance(int trigPin, int echoPin) {
13   digitalWrite(trigPin, LOW);
14   delayMicroseconds(2);
15   digitalWrite(trigPin, HIGH);
```

The Serial Monitor shows the output of the function, which is 'D:2' for each sensor.

Worklog:

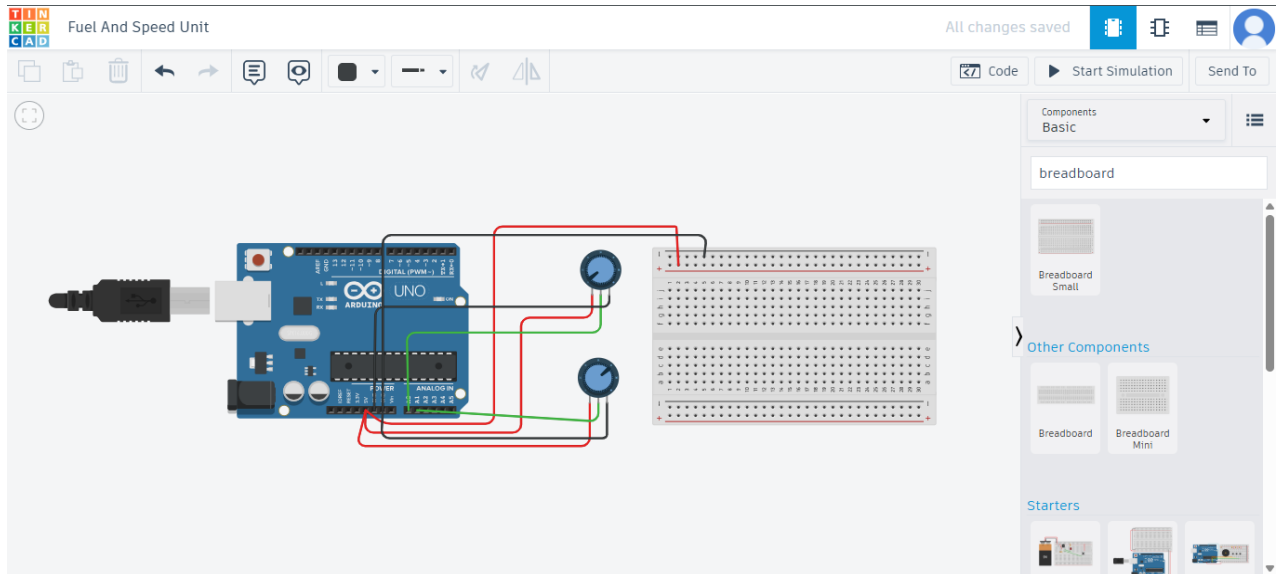
- Date: May 2, 2025
- On the project: Distance Sensor Unit
- Used Tinkercad to simulate four ultrasonic sensors.
- The TRIG and ECHO pins (pins 5–12) were connected to the Arduino.
- To determine the minimal distance, I wrote code for Arduino.
- The output was tested on the serial monitor (D:14, D:2, and D:30). The circuit and code were saved.
- Spending time: 1.5 hours

UNIT 2 – Fuel and Speed Sensor Unit :

Two potentiometers coupled to an Arduino UNO are used in this Fuel and Speed Sensor Unit configuration for reproducing real-time fuel and speed data. The first potentiometer serves as a sensor for the gasoline level, and the second one mimics the speed of the car. The voltage values transmitted to the microcontroller can be adjusted manually by adjusting the settings of the two potentiometers, which are connected to the Arduino's analog input pins (A0 and A1). The Arduino interprets these fluctuating voltages as indicating various acceleration and fuel levels. In order to replicate real-world logic, where higher speed results in higher fuel consumption, the system can be configured to indicate a falling fuel level when the speed potentiometer is turned to imitate higher speed.

Understanding the reading, processing, and potential applications of speed and fuel levels in real-world car systems is made very easy by this interactive simulation. Additionally, this unit's output can be routed to a buzzer or display in the Display Unit for aural and visual alerts. This circuit effectively illustrates the use of analog inputs to mimic sensor data for smart car monitoring.

Circuit Diagram:



Output:

The screenshot shows the Tinkercad code editor for 'Fuel And Speed Unit'. The code is as follows:

```
1 int fuelPin = A0;
2 int speedPin = A1;
3 float fuelLevel = 100.0; // Start with 100% fuel
4
5 void setup() {
6   Serial.begin(9600);
7 }
```

The serial monitor shows the following output:

```
Speed: 43 km/h Fuel Level: 71.55 %
Speed: 43 km/h Fuel Level: 71.12 %
Speed: 43 km/h Fuel Level: 70.69 %
Speed: 4 km/h Fuel Level: 70.65 %
Speed: 4 km/h Fuel Level: 70.61 %
Speed: 0 km/h Fuel Level: 70.61 %
Speed: 0 km/h Fuel Level: 70.61 %
Speed: 0 km/h Fuel Level: 70.61 %
Speed: 21 km/h Fuel Level: 70.40 %
Speed: 38 km/h Fuel Level: 70.02 %
Speed: 38 km/h Fuel Level: 69.64 %
Speed: 38 km/h Fuel Level: 69.26 %
Speed: 38 km/h Fuel Level: 68.88 %
Speed: 38 km/h Fuel Level: 68.50 %
Speed: 38 km/h Fuel Level: 68.12 %
Speed: 38 km/h Fuel Level: 67.74 %
```

Worklog:

- Date: 03-May-2025
- Worked on : speed and fuel sensor unit.
- Used Tinkercad to simulate two potentiometers to symbolize fuel and speed sensors.
- Potentiometers were connected to the Arduino analog pins A0 (fuel) and A1 (speed).

- Mapped speed input (0–120 km/h) and simulated fuel usage using Arduino code.
- Using Serial Monitor, the output was tested; dynamic fuel and speed readings were displayed.
- Potentiometer was adjusted to monitor real-time fuel percentage changes based on speed.
- Finished writing the code and saved the circuit.
- Spending time: 2 hours

UNIT 3 – Display Unit :

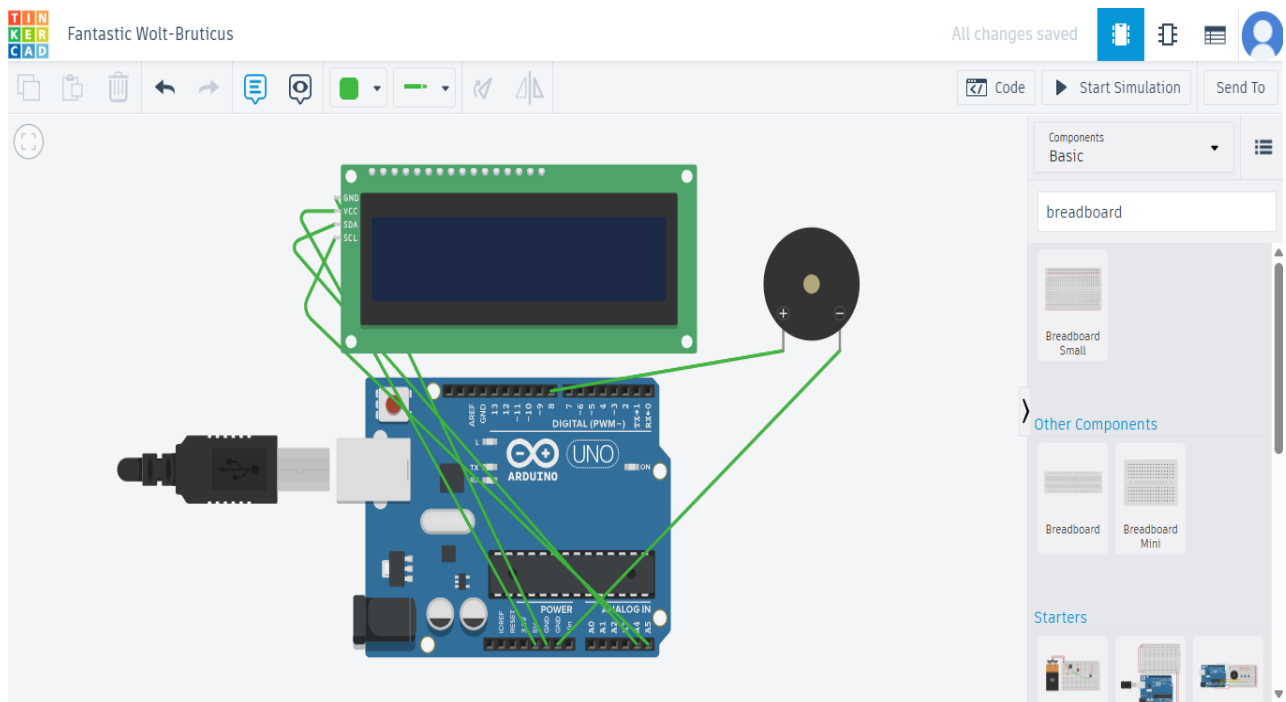
The primary objective of this Display Unit assignment was to acquire distance readings from a sensor and to offer both visual and aural notifications in the event that an obstruction was detected in the vicinity. An Arduino Uno microcontroller was linked to an ultrasonic sensor in order to accomplish this. Through the use of ultrasonic pulses and the calculation of the echo's return time, the ultrasonic sensor continuously determines the distance to the closest object. This time information is read by the Arduino, which uses a common formula to translate it into distance. The user can visually track the distance of an obstacle after it has been calculated and presented on an LCD screen.

When the obstacle is safely out of reach, like more than 20 cm, the system just displays the distance on the screen. But when the object approaches—into a harmful range—a warning message like "Warning! Obstacle" appears on the screen. In addition, a buzzer is turned on to provide an auditory warning. The buzzer doesn't always sound the same; it changes according to how close the object is. The beeping sound gets faster and more frequent the closer the obstacle is. In order to make the system more urgent as the object gets closer, the beeping intervals are changed by mapping the observed distance to the interval between buzzer tones.

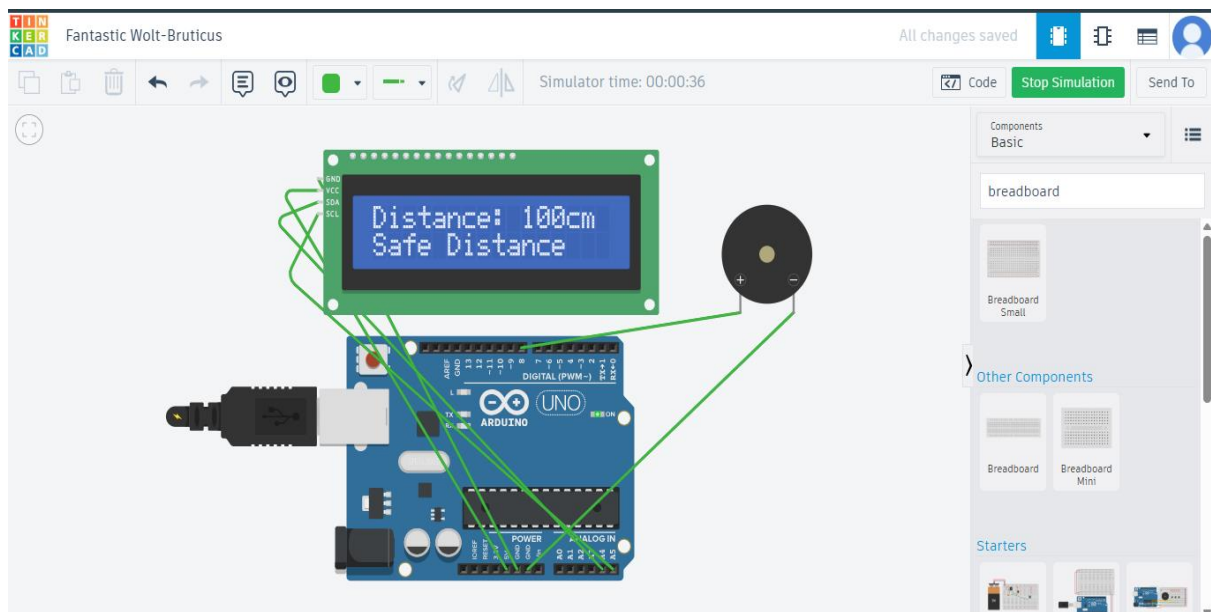
A real-time, dual-mode alarm system—visual and audio—is provided by this configuration, which aids in identifying surrounding

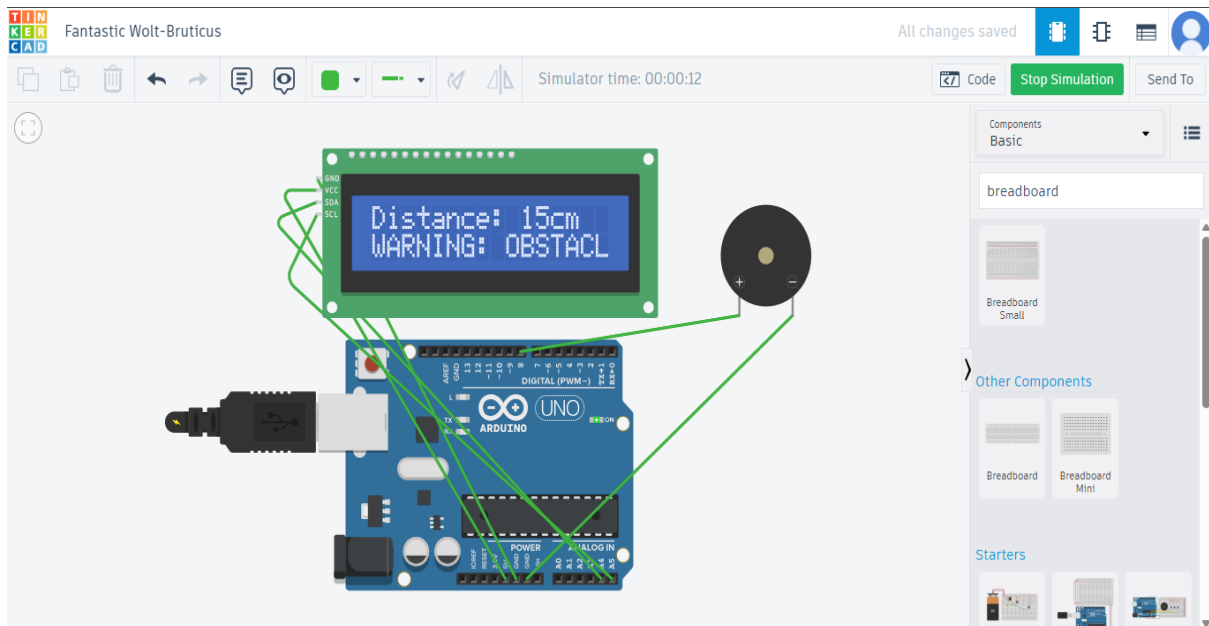
impediments and enabling prompt action. Robots, car parking systems, and any other system that needs safe navigation in real-world settings will find it especially helpful. By alerting users to possible collisions depending on distance closeness, this unit guarantees system safety.

Circuit Diagram :



Output:





WorkLog :

- Date: May 3, 2025
- Finished: Display Unit
- I2C LCD was interfaced with Arduino using theLiquidCrystal_I2C library.
- Arduino with LCD (I2C is) connected:
VCC → 5V
GND → GND
SDA to A4 SCL to A5 .
- Buzzer was connected to pin 8.
- Created Arduino code to use Serial (RS232) to receive distance values.
- Distance and messages were shown on the LCD screen.
- Buzzer warning that was activated based on distance and had a dynamic tone delay output that has been tested in both dangerous (<20 cm) and safe (>20 cm) situations.
- Checked the buzzer and LCD for correct output. Code and circuit saved.
- Duration: 1.5 hours